CS4151 Report

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1. Introduction

Prior to this module, I had no experience in modelling or animation, therefore for my project I decided to attempt to model and rig my own unique character as opposed to using a pre-made and pre-rigged model. I decided to go for a low poly character design that suited my experience level in this area. For my backdrop I decided to go for an old cartoon effect, bright magnetic colours and a low-resolution sky backdrop. The advanced techniques that are included while making this project are as follows: modelling & rigging my own model, animated camera, orientating the camera, and manipulating of the mesh.

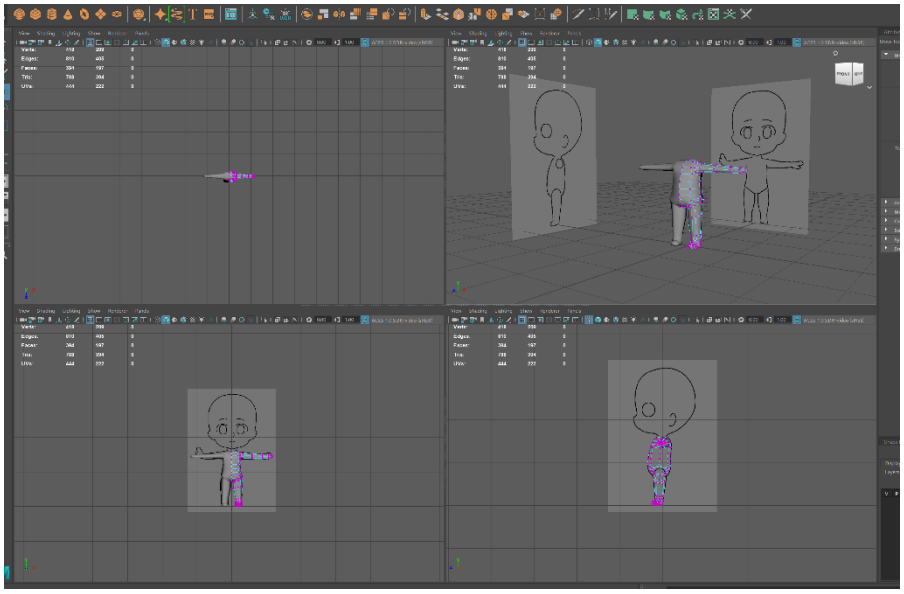
2. Resources

I download a front and side reference image from the internet, I did not follow the reference images exactly, but I had them imported in the Maya scene just to give me a general idea and reference on what shape to try and go for. I also used YouTube for a tutorial on how to properly rig the model which I followed as best as I could and feel that I did a good job. Both of these resources allowed me to model and rig my own character, despite having no prior experience in this area when starting this semester.

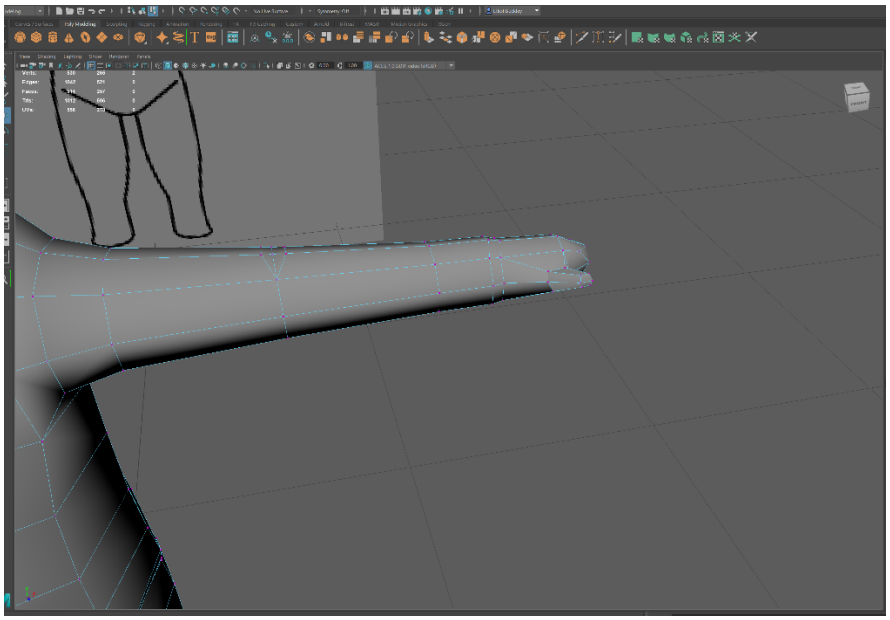
For my backdrop, I created the ground and trees using different polygons and altering the colour of the Lambert. No outside resources for were used for the physical backdrop mesh. As for the image, I created this is Microsoft Paint 3D, I wanted to go for a cheesy cartoon look and feel and I am happy with the way that this turned out, just using the paint bucket and blurred brush to add a bit of depth.

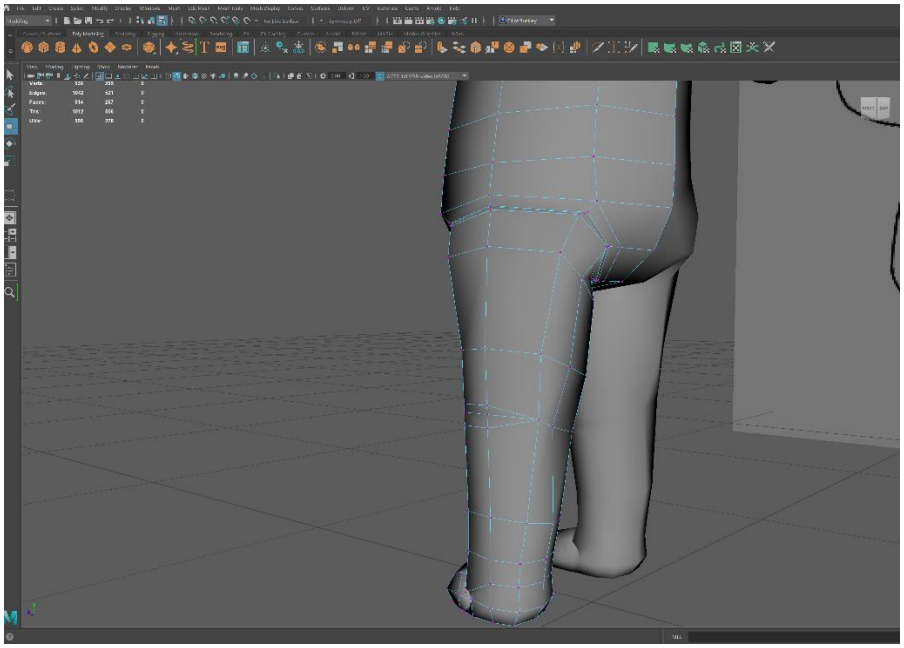
3. Process

I started off using a cube polygon and just adding new edges in order to create new faces to extrude. I used Maya's symmetry mode and only worked on the left side of the model, this made sure that everything was symmetrical.



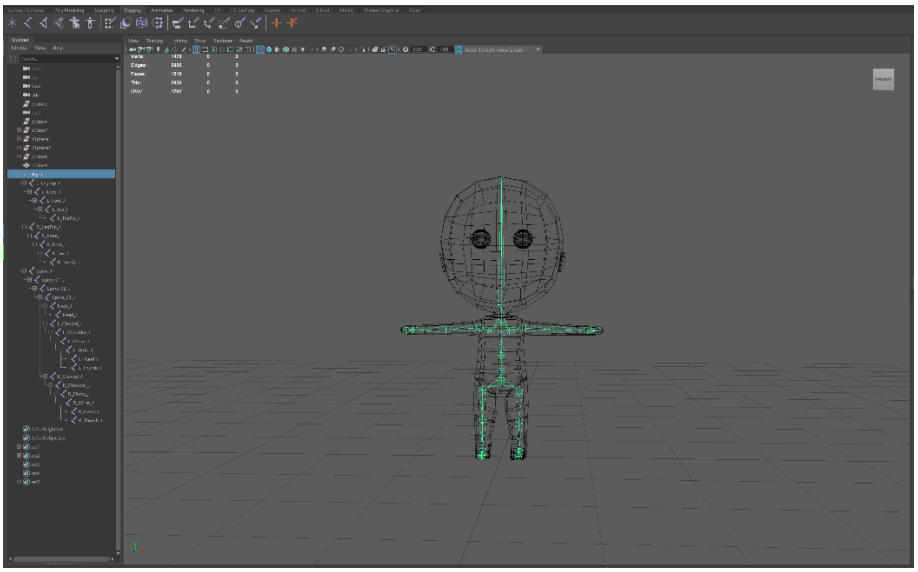
From there then I started working on the thumbs, feet, shaping the areas where the joints will be, such as the elbows, knees and where the neck was going to go.



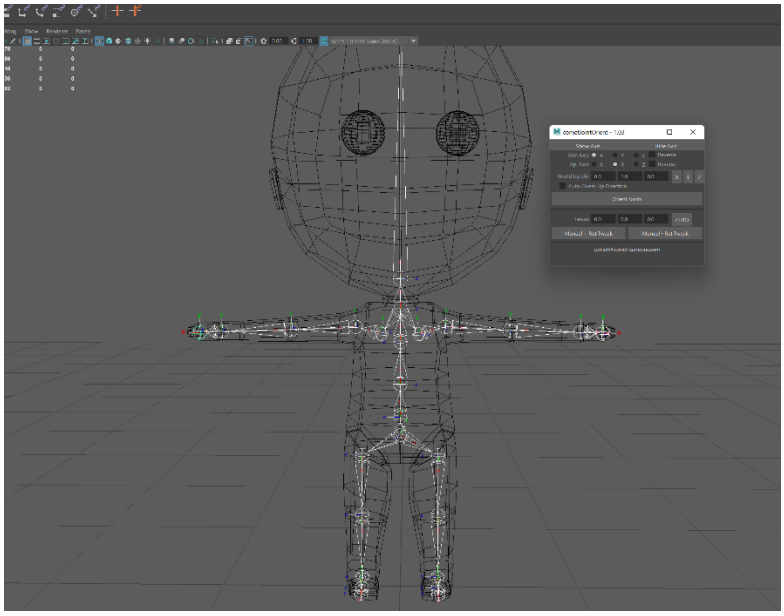


I created another polygon cube, to be used as the head and got to work shaping that to roughly match the reference images, I used another cube for the ears and a sphere for the eyes. Once everything was shaped to my satisfaction, I deleted the entire right side of the model and mirrored the left side, just to make sure everything was correct.

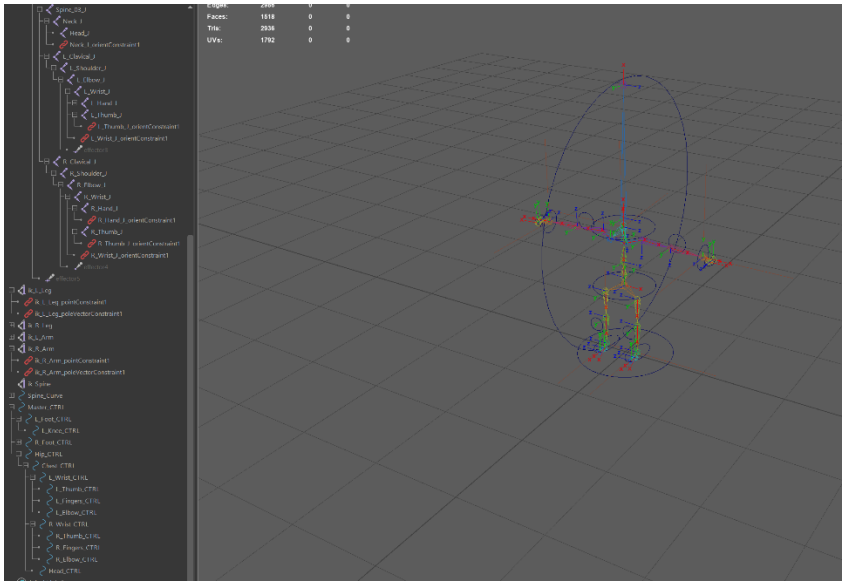
From here I got to work on the rigging, creating all of the joints, knees, ankles, hips, spine, arms, etc. This was the easiest part of the rigging process.



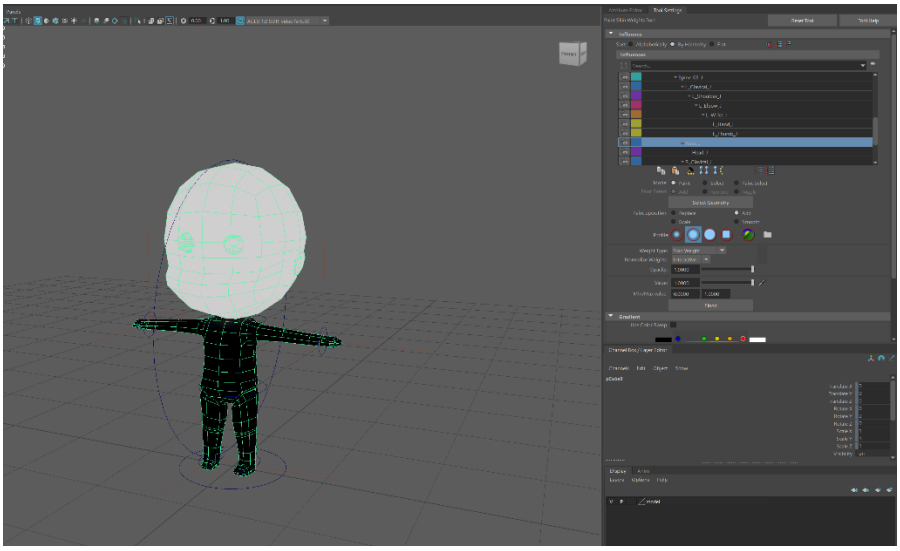
Next up I had to orientate all of the joints to ensure that the correct axis of each joint was facing the correct way, this was to ensure that they bent and rotated correctly when moving all of the limbs.



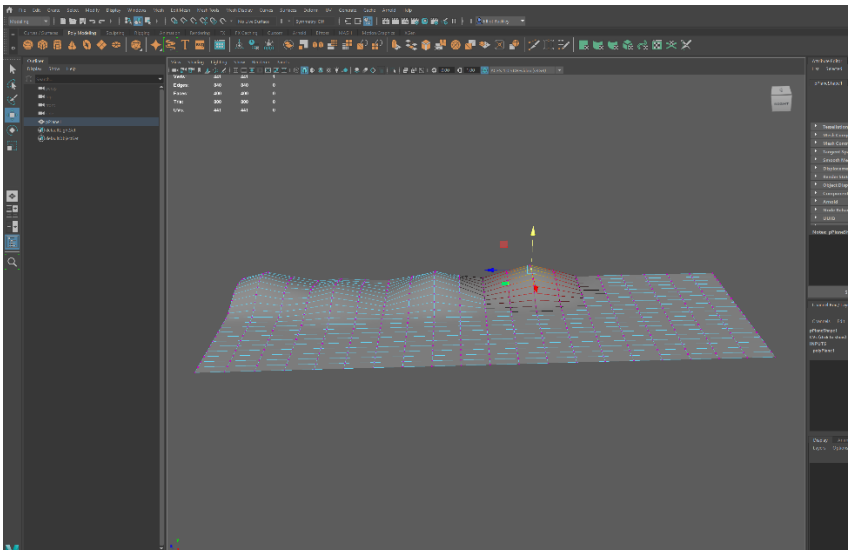
Next up I added controls to all of the major joints, the feet, wrists, neck, hips. This allowed me to apply animations to the controls and alter the controls position, as opposed to the joints, meaning if I moved the control on the left foot, the left hip and knee with move and rotate also, without altering their position.

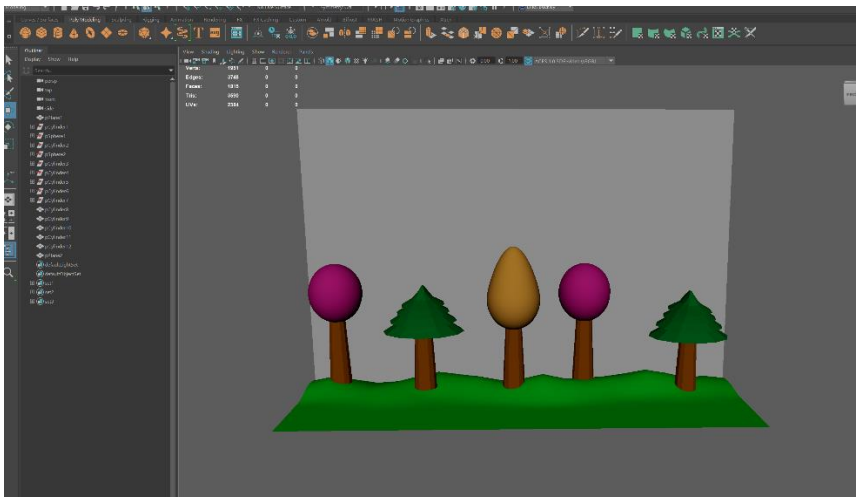


Lastly, for the actual model, I had to alter the weight of some of the joints, for example, the wrist joint was originally affecting the movement of the head. Maya has a tool that allows you to select a joint, see what areas it is affecting and add or removed parts of the model to be altered by said joint, and also how much of an affect the joint has on that area.

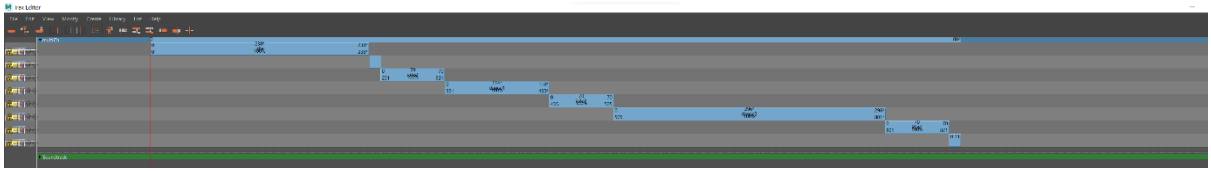


For the backdrop I altered the mesh of a polygon plane in order to give the effect of the terrain having hills, and also creating trees using polygon cylinders.

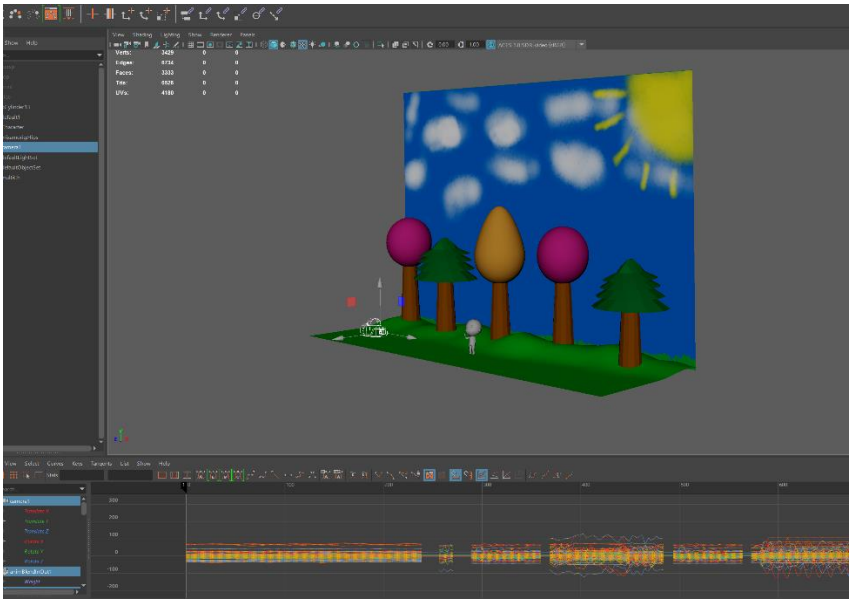




For the animation then, I had to create separate animation files for each of the animations I got from Mixamo, and then blend them into one long animation using the Trax Editor in Maya.



Lastly, I put everything into one scene and set up an animated camera to pan and zoom in on the character and backdrop as the animation played.



4. Reflection

If I were to do this project again, I would probably try to give my model more detail instead of going down the low poly route. I feel as though the animations would look much better, if the model had more joints, which it would if it had more detail. I would also try and actually give the model a UV layer, while I am proud of the end result of this project I feel as though it could look much better if the model was not just a grey blob.

Lastly, if I were to do this project again, I would try and actually create the animations myself as opposed to downloading them off the internet. While I did model and rig the character myself, I chose not to do the animations myself due to time constraints on my part. I would also like to include some more of the advanced techniques outlined in the project guidelines, despite doing what was specified I would have liked to maybe include some extra ones.

5. Appendix

I used a front and side view images which I got from the internet as a reference for my characters shape. For the rigging I had to watch a YouTube tutorial to be able to do it correctly without any clipping. Despite modelling and rigging the model myself I did also use Mixamo to create the animations for my character. YouTube was a very useful resource that I found myself using quite a bit to solve minor issues such as my model clipping with itself, correctly orientating joints on the rig, and just learning how to use Maya efficiently and all the different tools and components that it has.